LIFELINE Iteration 1: Approach Note (Simple Version)

1. How the data is made

Sizes are random within each tier's range. Crate value is roughly weight x (2 to 6) plus noise, so value per kg varies and a greedy pick can fail. Each case study's special features are forced with exact counts, not left to luck.

Case study	What the data looks like	Story reason
1 Delta Flood	Well-connected roads; 40-50% of locations have penalty 2+; 3-5 flood waves; mostly food and water crates	Flat delta, roads flood in waves
2 Mountain Earthquake	Very few roads (almost a tree); mostly medicine and tools; urgent alarms come in bursts	Few alternative routes; aftershocks trigger bursts
3 Coastal Cyclone	28-40% shelters; short, heavily overlapping time windows; noisy radio with many symbols; 3-5 waves	Short windows, noisy radio, surge closes roads
4 Urban Flash Flood	Dense roads; 1.5-1.8 alarms per tick; 10%+ rare crates	Many routes, calls come faster than service
5 Landslide Corridor	One long path-shaped road network, HQ at one end; one key road always cut	One highway through a gorge; a collapse isolates communities
6 Outbreak Camp	Mostly medicine and water; 55%+ of crates expire by hour 300; radio log repeats messages with one dominant symbol	Supplies spoil; beacons repeat

2. Hidden answers and how they are checked

Item	How it is built
Tour	Small tier: random order. Larger tiers: nearest neighbour, then 2-opt to improve it
Cover	Pick k nodes so every edge touches one of them, then link everything else to them
Safety rules	Start with a hidden valid assignment, flip a literal in any rule it breaks
Rations	Pick a subset and set the target to its sum
Patterns	Real log substrings (must be found) and random 40-symbol strings (must not be found)
Reports	report_b is an edited copy of report_a; edit_bound is the exact edit distance
Claims (small tier)	Generator solves with DP, Held-Karp and bitmasks; validator re-solves with brute force; both must agree

3. The traps

Trap	Setup	Why it hurts
46 Dock order	1500-3000 crates, fully sorted	Sorted input is the worst case for first-element-pivot quicksort
47 Best-looking crate	A few "bait" crates look best per kg but leave wasted space	Greedy reaches only about 56-60% of the optimum
48 Shortest looks cheapest	Chain of diamonds: one slow direct road or a fast two-road detour	Fewest-roads (BFS) costs at least 2x the cheapest route
49 Near miss	50k-100k symbols, nearly all the same; patterns match almost to the end, then fail	Naive search wastes about m comparisons at nearly every position

4. Where each case study gets hard

Case study	Hard part
1 Delta	Fewest roads and cheapest route differ; food/water crates have many near-ties

Case study	Hard part
2 Mountain	One wave can cut the valley, so the cable tree only covers the HQ side; bursts cause backlog spikes
3 Coastal	Max count and max value windows differ; near-uniform symbols mean Huffman barely beats 6 bits
4 Urban	Backlog grows steadily, so the data structure matters; many equal-cost routes make ties matter
5 Landslide	A target can be cut off (no route, partial tree); deep path breaks deep recursion
6 Outbreak	Value and expiry pull against capacity; repeated messages stress overlapping phrase search